Shiming Wu

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Education

The University of British Columbia, Vancouver, Canada

PhD in Economics	2020 - 2026 (expected)
Master of Science in Business Administration in Finance	2018 - 2020

Nankai University, Tianjin, China

Bachelor of Economics in Finance	2012 - 2015

Fields

Primary fields: Insurance, Financial Economics, Industrial Organization
Secondary fields: Public Economics (Environment & Health), Machine Learning

References

Paul Schrimpf
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Gorkem Bostanci
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Katherine Wagner
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David Green
david.green@ubc.ca

Prof. Gorkem Bostanci and Prof. Katherine Wagner have requested that our department submit letters for them. Please email econ.grad@ubc.ca for reference letters. Thanks.

Research Grants, Awards, and Fellowships

CIDER Small Grants in Innovative Data (C$26,000)	2023 – 2024
Sir Quo-Wei Lee Fellowship from St. John's College	2024 - 2026
Harnan A.N. Singh Scholarship in Economics	2022
President's Academic Excellence Initiative PhD Award	2020 – 2026
Faculty of Arts Graduate Fellowship	2020 – 2025
International Student Tuition Award	2018 – 2026
Merit Student and Scholarships from Nankai University	2012 - 2015

Job Market Paper

What Constrains Insolvency in Property Insurance?
Market Discipline, Capital Regulation, and Catastrophe Exposure

- This paper disentangles the roles of capital regulation and credit ratings in mitigating insolvency risk in the U.S. property insurance market. I first investigate the mechanism through which capital requirements affect the insurance market. Using reduced-form evidence and an instrumental variable approach that exploits a 2017 policy change as a quasi-experiment, I find that a $1 million increase in required capital leads insurers to hold $3.34 million more in capital and to raise insurance prices by 0.218 percentage points. These results reveal a direct trade-off between financial stability and consumer affordability. To further explore the underlying mechanisms, I develop a structural model in which insurers make capital and pricing decisions in a competitive market with limited liability and exposure to catastrophic risks. Counterfactual analyses show that tightening capital requirements improves solvency but raises prices. In the absence of capital regulation, the model predicts that the insolvency rate would increase by 0.09 percentage points, while insurance prices would decline by about 5.1%, accompanied by greater risk-taking and market concentration. A third counterfactual scenario examines a market without capital regulation but with high credit rating salience. When consumers place greater emphasis on credit ratings, the insolvency rate decreases; however, intensified price competition reduces profitability and increases market concentration. Overall, the findings underscore that capital regulation remains crucial for sustaining market stability, as heightened rating salience alone cannot fully substitute for its stabilizing effects.

Working Papers

Transformation of College to University and Impacts on Labor Market Outcomes and Training Participation under Employment Insurance

- This paper studies the impacts of different types of post-secondary education on education and labor market outcomes, with a particular focus on training participation during unemployment under Canada's Employment Insurance (EI) system. I first use variations of distances to institutions and exogenous variations of colleges that upgraded into universities to investigate the value added to university education. Then I separately estimate the impacts of the transformed universities and traditional colleges and universities. I compare results from OLS and IV regressions. To address treatment heterogeneity, I adapt the locally linear specification from Mountjoy (2022). Results suggest that the difference between university and college graduates is marginal. However, university entry improves labor market outcomes, such as employment and earnings, compared with cohorts without post-secondary education. Graduates from transformed universities obtain a higher probability of being employed and higher earnings compared with people without post-secondary education. Nonetheless, transformed university graduates may have worse performance in the labor market compared with graduates from colleges or traditional universities. In addition, transformed university graduates are more likely to register for EI-supported training programs during unemployment, compared with graduates from colleges, traditional universities, and individuals without post-secondary education, highlighting an important interaction between higher education pathways and training incentives embedded in the EI system.

The Impacts of Employer Health Insurance Mandates on Labour Market and Health: Evidence from the ACA

- This paper examines the effects of the Affordable Care Act's employer mandate on labor market outcomes and health. I exploit variation in the staggered implementation of the employer mandate using a staggered difference-in-differences research design. The main analysis employs a doubly robust difference-in-differences estimator to mitigate selection bias, while honest difference-in-differences methods are used to assess the robustness of key results. The full-sample results indicate that the employer mandate did not significantly increase employer-sponsored health insurance coverage or generate meaningful changes in overall labor market outcomes. However, the effects are heterogeneous across income groups. Among low-income workers, the mandate led to a 9.31 percentage points increase in employer-sponsored health insurance coverage, which was associated with improved self-reported health and greater utilization of health care services. At the same time, low-income workers became more vulnerable to involuntary part-time employment, suggesting that employers were more likely to reduce work hours than to reduce employment in response to the mandate. These findings imply that the employer mandate may improve worker productivity over the long run through better health, but it may also exacerbate income inequality by disproportionately exposing low-income workers to reductions in work hours.

Contracting the Cosmos: Externalities, Capital Markets, and Coasian Bargaining

- The growing energy requirements of artificial intelligence (AI) present significant challenges for terrestrial power grids, prompting the technology sector to consider deploying data centers in low Earth orbit. However, operating in the orbital commons introduces an unpriced spatial externality: the endogenous jump-to-ruin risk associated with orbital congestion, known as the Kessler Syndrome. This paper develops a continuous-time stochastic real options game featuring endogenous asset loss and exogenous technological shocks to examine the capital structure and entry timing for commercial space infrastructure. The analysis highlights a theoretical tension: in environments subject to endogenous jump-to-ruin, standard uncoordinated competitive entry increases physical asset density, thereby elevating the probability of collision and potential market failure. Furthermore, the model shows that traditional debt financing is structurally unviable due to zero collateral recovery upon collision, suggesting a pure equity structure. To address the spatial externality, I model two equilibrium regimes. Under a mandatory insurance regime, global reinsurance syndicates function as decentralized regulatory mechanisms, pricing the spatial externality into a firm's cost of capital. In a self-insured open access regime, physical carrying capacity constraints and endogenous debris removal costs incentivize Coasian bargaining. The findings suggest that the unregulated space market tends to consolidate into a coordinated oligopoly, mitigating the tragedy of the commons through private financial constraints without requiring direct sovereign intervention.

A Reinforcement Learning Approach to Dynamic Capital Regulation in Property Insurance

- Regulators in the U.S. property insurance market face a critical challenge: transitioning from static, formula-based capital requirements to dynamic, model-based regimes. While dynamic regulation offers the potential to improve social welfare and market resilience, it suffers from two major barriers: computational intractability under profound uncertainty and a lack of interpretability required for regulatory oversight. In this paper, I propose a novel framework for outcome-based regulatory design. I develop a "Learn-Verify-Explain" methodology that utilizes Deep Reinforcement Learning to discover optimal dynamic capital strategies. Unlike traditional black-box approaches, my framework integrates Formal Verification to mathematically guarantee compliance with safety constraints and Decision Tree Extraction to distill complex policies into transparent, implementable rules. Empirical results demonstrate that this hybrid approach outperforms traditional static benchmarks, increasing social welfare by approximately 35% while reducing insolvency rates to zero. Crucially, the distilled policy reveals a risk-sensitive stabilization strategy: the agent learns to prioritize market efficiency through deregulation during stable periods, while imposing immediate corrective tightening upon detecting early signs of distress. This study provides an experiment of "AI-in-the-loop" financial regulation.

Adaptation, Insurer Solvency, and Affordability

- I analyze the growing tension between insurer solvency and insurance affordability in property markets exposed to climate change. I develop a structural model where risk-averse households choose between purchasing insurance and investing in physical risk adaptation, while competitive insurers face regulatory capital requirements for systemic climate risk. My model demonstrates two findings: (1) high frictional cost of regulatory capital creates a high entry price for insurance that is relatively insensitive to individual adaptation, and (2) below a specific wealth threshold, both physical adaptation and insurance coverage collapse as households lack the liquidity to clear the solvency-loaded premium. Through policy counterfactuals, I find that premium subsidies are counterproductive; by making insurance artificially cheap, they cause households to increase financial coverage while decreasing physical adaptation effort. In contrast, adaptation grants that directly target the liquidity friction are highly effective, allowing households to jump to a higher state of resilience, which improves both household welfare and market-wide financial stability.

From Monoculture to Market: The Economics of Plant Biodiversity, Systemic Risk, and Consumer Welfare

- Modern industrial agriculture relies on monocultures to maximize economies of scale, yet this genetic homogenization introduces profound systemic risks to global food security. This paper investigates whether market competition naturally corrects this trend or exacerbates it, utilizing a two-stage structural model of endogenous cultivar choice and price competition. By estimating a random coefficients Logit model, I quantify the static welfare gains from cultivar variety and the dynamic insurance value of biodiversity against correlated supply shocks. Simulation results demonstrate that consumers derive significant utility from variety, with a transition to pure monoculture resulting in a 73% loss in static consumer surplus. Crucially, I find that market structure determines this vulnerability: a simulated competitive oligopoly sustains four distinct cultivars, whereas a monopoly restricts the market to just two, internalizing cannibalization. Furthermore, I identify a substantial market failure: while monoculture is statically efficient in normal states, it creates unpriced systemic tail risk. Under simulated shocks, the expected social welfare of a diverse biological portfolio is approximately 68% higher than that of a monoculture. These results suggest that profit-maximizing firms structurally under-provide biodiversity, providing a rigorous economic rationale for biodiversity subsidies and stringent antitrust scrutiny in agribusiness.

Defensive Triage: Minimizing Insurance Liability via Bayesian Uncertainty Quantification in Emergency Queues

- Emergency Department (ED) overcrowding presents a dual crisis: it compromises patient safety and destabilizes the medical liability insurance market. High-cost malpractice claims, particularly those stemming from failure to diagnose ambiguous cases, drive up insurance premiums, thereby increasing healthcare overhead and limiting patient access to affordable care. Traditional triage protocols and deterministic Machine Learning (ML) models optimize for operational efficiency but ignore the fat-tail liability risks associated with diagnostic uncertainty. This paper proposes a novel "Defensive Triage" mechanism that integrates Bayesian Uncertainty Quantification into queueing logic. By prioritizing patients based on epistemic uncertainty, we effectively hedge against catastrophic misdiagnosis. Using a discrete-event simulation benchmarked against standard protocols, I demonstrate that this approach significantly reduces aggregate liability exposure. I argue that by lowering the frequency of high-severity claims, Defensive Triage allows insurers to stabilize premiums. This creates a virtuous economic cycle: reduced insurer risk lowers provider costs, ultimately expanding the capacity of the healthcare system to provide accessible, insured care to a broader population.

Hedging Against the Silver Tsunami: A Health Capital Preservation Triage Framework to Mitigate Long-Term Care Liability and Insurance Risks

- As populations age, the primary economic threat to healthcare sustainability is the accelerated depreciation of health capital. While traditional triage focuses on short-term liability shielding, musculoskeletal (MSK) triage in elderly populations requires a long-term capital preservation approach. This research proposes three novel frameworks: The Functional Autonomy Hedge (FAH), which utilizes gradient-based symptom volatility; Expected Liability Calibration (ELC), which optimizes for insurance solvency by pricing the marginal cost of diagnostic delay; and the Frailty-Decay Shield (FDS), which incorporates a biological aging parameter. By identifying latent invisible risks near clinical cliffs, I employ actuarial policies to prioritize patients with high-risk profiles. Simulation results ($N = 5,000$) demonstrate that while FAH offers a modest 5.6% liability reduction over standard AI, the optimized ELC framework achieves a 50.2% reduction. The FDS framework, by pricing frailty, provides the superior result with a 52.4% reduction, effectively serving as a mathematically rigorous hedge against the systemic risks of the Silver Tsunami.

Work in Progress

The Energy Bridge: Optimal Sequencing of Terrestrial Nuclear and Space-Based Power for AI Computation

- The rapid expansion of artificial intelligence is creating a terminal energy bottleneck that traditional terrestrial grids are ill-equipped to handle. While space-based solar power represents the ultimate solution for high-density compute clusters, the technology remains in its infancy, necessitating a multi-decade transition strategy involving terrestrial solar and Small Modular Reactors (SMRs). This paper develops a structural corporate finance model to analyze the optimal sequencing of these heterogeneous energy investments under conditions of technological uncertainty and capital irreversibility. Using a continuous-time real options framework, I examine the trade-offs between the lower capital intensity of terrestrial solar and the high-baseload reliability of nuclear SMRs as bridges to eventual orbital energy migration. The model characterizes the technological overhang created by long-lived nuclear assets and the resulting stranded asset risk if space-based breakthroughs occur prematurely. I solve the firm's intertemporal optimization problem to derive endogenous investment thresholds and optimal capital structure dynamics. The paper aims to propose a novel class of contingent-claim financing instruments. The resulting framework provides a rigorous valuation architecture for technology firms and infrastructure investors navigating the high-stakes frontier of the AI energy transition.

Teaching

ECON 502 Macroeconomics, teaching TA (UBC)
ECON 490 Seminar in Applied Economics, TA (UBC)
ECON 370 Benefit-Cost Analysis and the Economics of Project Evaluation, TA (UBC)
ECON 356 Introduction to International Finance, TA (UBC)
ECON 345 Money and Banking, TA (UBC)
ECON 255 Understanding Globalization, teaching TA (UBC)
ECON 102 Principles of Macroeconomics, teaching TA (UBC)
ECON 101 Principles of Microeconomics, teaching TA (UBC)

Conferences

2024: 19th CIREQ PhD Students' Conference
2022: NBER Insurance Working Group Meeting

Employment

Data analyst at BC ACARN, University of British Columbia	2026 - present, Vancouver, Canada
Teaching assistant at University of British Columbia	2021 - present, Vancouver, Canada
Research assistant at University of British Columbia	2019 - 2024, Vancouver, Canada
Tutor and teaching assistant at Yingde Inspiration Education	2017 - 2018, Yingde, China
Auditor at KPMG	2015 - 2016, Beijing, China

Other Information

Computer Skills	R, Julia, Stata, MATLAB, SPSS, Python, LaTeX, SQL, C++
Data Science	Machine learning, Causal inference, Structural estimation, A/B testing
Mathematics	Theoretical modeling, Stochastic optimization & optimal control
	Reinforcement learning, Game theory & mechanism design
System Architecture	Linux(Ubuntu), Custom PC architecture & integration
Languages	English(fluent), Mandarin(native), Cantonese(native), French(learning), German(learning)
Certificates	Passed CFA Level II Exam
Datasets	Bloomberg, Capital IQ, Large-scale administrative datasets
Gender	Female